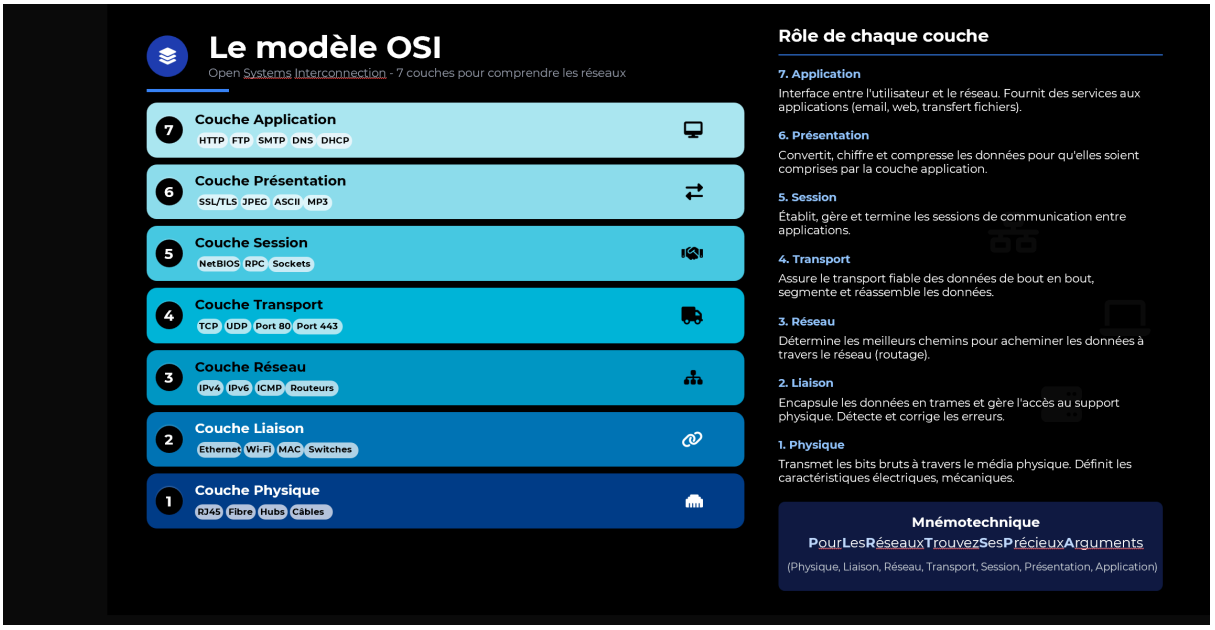
DOCUMENTATION TP2



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1. Connexion et préparation du serveur

Connexion au serveur Windows en SSH avec le compte **Administrateur**.

L'objectif est de désactiver le DHCP automatique sur l'interface réseau afin de configurer une adresse IP statique.

Affichage des interfaces réseau

```
Get-NetIPInterface
```

Désactivation du DHCP sur l'interface cible

```
Set-NetIPInterface -InterfaceIndex 6 -Dhcp Disabled
```

2. Configuration IP statique

Une adresse IP fixe est attribuée au serveur afin d'assurer la stabilité des services DNS et DHCP.

```
New-NetIPAddress -InterfaceIndex 6 -IPAddress 192.168.100.1
0 -PrefixLength 24 -DefaultGateway 192.168.100.1
```

Suppression possible si erreur de configuration :

```
Remove-NetIPAddress -InterfaceIndex 6 -IPAddress 192.168.10
0.10 -PrefixLength 24 -DefaultGateway 192.168.100.1
```

Configuration DNS statique

```
Set-DnsClientServerAddress -InterfaceIndex 6 -ServerAddress
es ("192.168.100.10","8.8.8.8")
```

3. Installation et configuration du rôle DNS

Installation du service DNS

```
Install-WindowsFeature DNS -IncludeManagementTools
```

```
Get-WindowsFeature D
```

```
Get-Service DNS
```

Ajout des redirecteurs DNS (Forwarders)

```
Add-DnsServerForwarder -IPAddress 8.8.8.8,8.8.4.4
```

4. Installation du rôle DHCP

```
Install-WindowsFeature -Name DHCP -IncludeManagementTools
```

```
Set-Service -Name DHCPServer -StartupType Automatic
```

```
Start-Service -Name DHCPServer
```

5. Vérification du nom complet du serveur (FQDN)

```
(Get-WmiObject Win32_ComputerSystem).DNSHostName + "." + (G
et-WmiObject Win32_ComputerSystem).Domain
```

Renommage du serveur

```
Rename-Computer -NewName "SRV-AD-DNS-DHCP" -Restart
```

6. Configuration des scopes DHCP

Chaque sous-réseau possède une étendue DHCP spécifique.

Scope USERS

```
Add-DhcpServerv4Scope -Name "Scope-USERS" -StartRange 192.1
68.10.2 -EndRange 192.168.10.254 -SubnetMask 255.255.255.0
-State Active
```

```
Set-DhcpServerv4OptionValue -ComputerName "SRV-AD-DNS-DHCP"
-ScopeId 192.168.10.0 -Router 192.168.10.1 -DnsServer 192.1
68.100.10 -DnsDomain "entreprise.lab"
```

Scope VIP

```
Add-DhcpServerv4Scope -Name "Scope-VIP" -StartRange 192.16
8.20.2 -EndRange 192.168.20.254 -SubnetMask 255.255.255.0 -
State Active
```

```
Set-DhcpServerv4OptionValue -ComputerName "SRV-AD-DNS-DHCP"
-ScopeId 192.168.20.0 -Router 192.168.20.1 -DnsServer 192.1
68.100.10 -DnsDomain "entreprise.lab"
```

Scope SRV

```
Add-DhcpServerv4Scope -Name "Scope-SRV" -StartRange 192.16
8.100.51 -EndRange 192.168.100.254 -SubnetMask 255.255.255.
0 -State Active
```

```
Set-DhcpServerv4OptionValue -ComputerName "SRV-AD-DNS-DHCP"
-ScopeId 192.168.100.0 -Router 192.168.100.1 -DnsServer 19
2.168.100.10 -DnsDomain "entreprise.lab"
```

Scope DMZ

```
Add-DhcpServerv4Scope -Name "Scope-DMZ" -StartRange 192.16
8.200.51 -EndRange 192.168.200.254 -SubnetMask 255.255.255.
0 -State Active
```

```
Set-DhcpServerv4OptionValue -ComputerName "SRV-AD-DNS-DHCP"
-ScopeId 192.168.200.0 -Router 192.168.200.1 -DnsServer 19
2.168.100.10 -DnsDomain "entreprise.lab"
```

7. Vérifications

```
Get-Service DHC
```

```
Get-DhcpServerv4Scope -ComputerName "SRV-AD-DNS-DHCP"
```

```
Get-DhcpServerv4OptionValue -ComputerName "SRV-AD-DNS-DHCP"
-ScopeId 192.168.10.0
```

8. Options avancées DHCP (optionnel)

Ajouter une exclusion d'adresse

```
Add-DhcpServerv4ExclusionRange -ScopeId 192.168.10.0 -Start
Range 192.168.10.1 -EndRange 192.168.10.10
```

Ajouter une réservation

```
Add-DhcpServerv4Reservation -ScopeId 192.168.100.0 -IPAddre
ss 192.168.100.13 -ClientId "08-00-27-CB-A4-2A" -Descriptio
n "SRV-SIEM"
```

```
Add-DhcpServerv4Reservation -ScopeId 192.168.100.0 -IPAddre
ss 192.168.100.12 -ClientId "08-00-27-20-44-92" -Descriptio
n "SRV-SMB"
```

Supprimer une réservation

```
Remove-DhcpServerv4Reservation -ScopeId 192.168.200.0 -Clie
ntId "08-00-27-02-AD-06"
```

9. Problème de réservation côté Linux

Modifier le fichier :

```
/etc/dhcp/dhclient.conf
```

Ajouter :

```
send dhcp-client-identifier = hardware;
```

Puis :

```
sudo dhclient -r enpos3
```

10. Configuration pfSense – DHCP Relay

Dans l'interface PfSense :

Service → DHCP Relay

Sélectionner les interfaces concernées

Ajouter le serveur DHCP :

```
192.168.100.10
```

Le routeur attribue automatiquement le scope selon le sous-réseau.

11. Autorisation du serveur DHCP dans Active Directory

Si problème de droits :

```
runas /user:ENTREPRISE\Administrateur powershell
```

Réinitialisation du mot de passe Administrateur :

```
Set-ADAccountPassword -Identity Administrateur -Reset -NewP
assword (ConvertTo-SecureString "Xazerty1" -AsPlainText -Fo
rce)
```

Autorisation du serveur DHCP :

```
Add-DhcpServerInDC -DnsName "entreprise.lab" -IPAddress 19
2.168.100.10
```

SCRIPT D'AUTOMATISATION

intégrer une création de réservation

```
Clear-Host

function Show-Menu {

Write-Host "===== AD DHCP DNS MENU =====" 
Write-Host "0 - Exit"
Write-Host "1 - Rename computer"
Write-Host "2 - Set Net IP Address"
Write-Host "3 - Install DNS"
Write-Host "4 - Install DHCP"
Write-Host "5 - Create DHCP Scope"

}

do {
Show-Menu
$choice = Read-Host "Your choice"

switch ($choice) {

    "0" {
        Write-Host "Exit..."
```

```
        break
    }

    "1" {
        $name = Read-Host "Enter new hostname"
        Rename-Computer -NewName $name -Restart
    }

    "2" {
        Get-NetAdapter
        $iface = Read-Host "Interface Alias"
        $ip = Read-Host "IP Address"
        $prefix = Read-Host "Prefix (ex 24)"
        $gw = Read-Host "Gateway"
        $dns = Read-Host "DNS Server"

        Get-NetIPAddress -InterfaceAlias $iface -AddressFam
ily IPv4 -ErrorAction SilentlyContinue |
        Remove-NetIPAddress -Confirm:$false -ErrorAction Si
lentlyContinue

        New-NetIPAddress -InterfaceAlias $iface -IPAddress
$ip -PrefixLength $prefix -DefaultGateway $gw
        Set-DnsClientServerAddress -InterfaceAlias $iface -
ServerAddresses $dns

        ipconfig

        # ===== MESSAGE DE SUCCÈS AJOUTÉ =====
        Write-Host "IP configuration SUCCESSFULLY applied."
-ForegroundColor Green
    }

    "3" {
        Install-WindowsFeature DNS -IncludeManagementTools
        $forward = Read-Host "Enter DNS Forwarder (ex 8.8.
8.8)"
        Add-DnsServerForwarder -IPAddress $forward
```

```
    }

    "4" {
        Install-WindowsFeature DHCP -IncludeManagementTools
        Set-Service DHCPServer -StartupType Automatic
        Start-Service DHCPServer
    }

    "5" {
        $scopename = Read-Host "Scope Name"
        $start = Read-Host "Start Range"
        $end = Read-Host "End Range"
        $mask = Read-Host "Subnet Mask"
        $scopeid = Read-Host "ScopeID (ex 192.168.10.0)"
        $router = Read-Host "Gateway"
        $dns = Read-Host "DNS Server"
        $domain = Read-Host "Domain"

        Add-DhcpServerv4Scope -Name $scopename -StartRange
$start -EndRange $end -SubnetMask $mask -State Active

        Set-DhcpServerv4OptionValue `
            -ScopeId $scopeid `
            -Router $router `
            -DnsServer $dns `
            -DnsDomain $domain
    }

    default {
        Write-Host "Invalid choice"
    }
}

Pause
} while ($choice -ne "0")
```